

world, Adam does not worry about health risks, disability, or potential involuntary job loss. He can continue work over the next thirty years earning a constant inflation-adjusted salary.

Adam retires at the start of the thirty-first year, and his retirement lasts thirty years. Withdrawals are made at the beginning of each year during retirement. The withdrawal amount is defined as the percentage of retirement date assets withdrawn (e.g., 4 percent), and this amount adjusts for inflation in subsequent years. The maximum sustainable withdrawal rate over thirty years is the initial percentage of assets withdrawn in the first year. That amount is adjusted for inflation in subsequent years, and the portfolio balance reaches zero at the end of the thirtieth year of retirement. For simplicity's sake, portfolio administrative and planning fees are not charged, and taxes are not deducted.

The historical characteristics for the 50/50 portfolio include a 5.6 percent real arithmetic mean, volatility of 10.7 percent, and a real compounded return of 5.1 percent. If the historical average compounded return could be fixed at 5.1 percent without volatility, a 15 percent savings rate for thirty years would result in retirement date wealth equal to 10.1 times salary. Exhibit 3.4 shows the historical wealth accumulations based on actual rolling periods of the historical data. Though Adam could expect (on average) a wealth accumulation equal to 10.1 times his salary, the historical outcomes ranged from a minimum of 5.2 times if he retired in 1982 to a maximum of 17.4 times if he retired in 2000. Despite saving in the same way from identical salaries and using the same 50/50 asset allocation, one Adam was able to save more than 3.3 times as much as the other. These are very different outcomes, again, for individuals who otherwise behaved in identical ways and were only exposed to a different sequence of market returns during their working years.

#### *Exhibit 3.4*

##### *Wealth Accumulation after a 30-Year Career*

##### *50/50 Asset Allocation, 15% Savings Rate, Inflation-Adjusted Salary*

##### *Using SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds*